Can a Fock hierarchy understand and reduce complex biochemical networks?

A rigorous evaluation of positive representations, modal couplings, memory, and classical algorithms

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Five reaction systems | Up to 279,936 microstates
130 configurations and numerical controls

Conclusion in one sentence
A multi-reference Fock basis with squared amplitudes compresses the probability distribution of coupled reaction networks surprisingly well, but does not outperform the best low-rank reductions tested in speed or maximum accuracy. The demonstrated advantage is in *representation*, not yet in a *scalable algorithm*.

Exploratory numerical research report.
10 November 2026. Results computed using accompanying Python code; no experimental biological data are used.

Executive summary

This study critically reassesses a Fock-space-inspired proposal for reducing stochastic dynamics: express a probability distribution using modes of different orders, evolve these modes with the network generator, and account for the effect of higher-order interactions on observables and transition probabilities. The chosen problem is the chemical master equation (CME), whose generator is exactly known once the reactions have been specified.

Five finite networks with between 2,401 and 279,936 states were compared against a full reference solver. Alternatives include linear Galerkin projection, squared amplitudes with one or several reference centers, an exponential probability family, state mixtures with a reduced Markov generator, closure with auxiliary memory variables, mean-field methods, finite-state projection, and dynamical matrix low-rank approximation (DLR). No reduced method is trained against the exact solution.

- **Main favorable result:** for a bistable five-species network (7,776 microstates), the multi-reference method yields $L^1 = 0.0326$ with 400 modes; rank-2 DLR obtains 0.0652 with 500 parameters. In a six-species cascade (46,656 states), Fock obtains 0.0476 with 245 modes, versus 0.0681 for rank-2 DLR with 860 parameters.
- **Strong unfavorable control:** rank-4 DLR achieves $L^1 = 0.0162$ for bistability, 0.00955 for the cascade, and 0.0200 for seven species, outperforming the tested Fock construction when a larger dimension is allowed.
- **Probability representation:** the ansatz $P_n \propto \Psi_n^2$ preserves positivity and normalization by construction. It does not establish exactness of the projected dynamics.
- **Unresolved scalability:** to evolve the amplitudes, $Q(P)$ is computed over the full state space. For seven species the 215-mode variant takes around 4.6 seconds, compared with 0.14 seconds for rank-2 DLR and 0.47 seconds for direct CME propagation on the tested machine.

Verdict

There is a mathematically interesting route toward compression via shifted bases and explicit positivity. Evidence is insufficient to call this theory a computational method superior to existing approaches. Further investigation is warranted only if modal evaluation can avoid enumerating all microstates while retaining the accuracy advantage on harder networks.

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1 Scientific question and falsifiable hypothesis

A Fock representation of a stochastic system with d species is natural: creation, destruction, and population changes can be expressed as operations between occupation states. But writing a problem in Fock space does not automatically reduce its dimensionality. If each species has L occupation levels, the state space contains L^d microstates.

The hypothesis tested here is more specific: a distribution can be propagated using a moderate number of modes around several relevant coherent reference states, preserving nonlinear interactions and transitions more accurately than classical approximations of comparable complexity. The verification criteria are:

1. A correct generator constructed solely from chemical reaction rates, without future solution information.
2. No negative probabilities, or explicit disclosure when a method produces them.
3. Errors on the entire distribution, not only the mean and variance.
4. Comparisons with the same number of coefficients and with the same computational budget.
5. Validation of joint probabilities rather than only marginal distributions.
6. Variation in the number of modes and numerical stability tests.

This is a deterministic benchmark on simulated reaction models, not experimental validation with cells or statistical testing across subjects. Attaching p-values to integration errors would not be appropriate.

2 From chemical reactions to a stochastic generator

2.1 Chemical master equation

Let the microstate be $\boldsymbol{n} = (n_1, \ldots, n_d)$ with $n_i \in \{0, \ldots, M\}$ and $M = L - 1$. For reaction r with stoichiometric jump $\boldsymbol{\nu}_r$ and propensity $a_r(\boldsymbol{n})$,

$$\frac{dP(\boldsymbol{n}, t)}{dt} = \sum_r \left[a_r(\boldsymbol{n} - \boldsymbol{\nu}_r) P(\boldsymbol{n} - \boldsymbol{\nu}_r, t) - a_r(\boldsymbol{n}) P(\boldsymbol{n}, t) \right]. \tag{1}$$

Terms involving out-of-domain states vanish. In matrix notation $\dot{\boldsymbol{p}} = Q\boldsymbol{p}$, where

$$Q_{\boldsymbol{m},\boldsymbol{n}} = \sum_r a_r(\boldsymbol{n}) \left[\mathbf{1}_{\boldsymbol{m}=\boldsymbol{n}+\boldsymbol{\nu}_r} - \mathbf{1}_{\boldsymbol{m}=\boldsymbol{n}} \right]. \tag{2}$$

For a valid CME, $Q_{ij} \geq 0$ for $i \neq j$ and $\mathbf{1}^\top Q = 0$. Hence $\exp(Qt)$ defines stochastic evolution: $\boldsymbol{p}(t) \geq 0$ and $\sum_{\boldsymbol{n}} P(\boldsymbol{n}, t) = 1$ whenever these conditions hold initially.

2.2 Reaction-network family used in the experiment

For species i, we include birth, self-activation, regulation by the preceding neighbor, and degradation:

$$\varnothing \xrightarrow{b_i(\boldsymbol{n})} X_i, \qquad X_i \xrightarrow{\mu_i n_i} \varnothing, \tag{3}$$

$$b_i(\boldsymbol{n}) = \left(1 - \frac{n_i}{M}\right) \left[b_i^{(0)} + u_i H(n_i) + v_i G_i(n_{i-1}) \right], \tag{4}$$

$$H(z) = \frac{z^q}{(M/2)^q + z^q}, \qquad G_i(z) = H(z) \text{ or } 1 - H(z). \tag{5}$$

The capacity factor suppresses production when $n_i = M$; degradation vanishes at $n_i = 0$. All parameters and initial conditions are defined in `experiment.py` and exported to JSON. The Hill function describes cooperative nonlinear coupling; these rates are not claimed to correspond to any particular biological protein.

3 Deriving the Fock-space structure

3.1 Occupation states and operators

In Doi–Peliti notation, define

$$|P(t)\rangle = \sum_{\boldsymbol{n}} P(\boldsymbol{n},t)|\boldsymbol{n}\rangle, \quad a_i^\dagger|\boldsymbol{n}\rangle = |\boldsymbol{n}+\boldsymbol{e}_i\rangle, \quad a_i|\boldsymbol{n}\rangle = n_i|\boldsymbol{n}-\boldsymbol{e}_i\rangle. \tag{6}$$

For birth–death processes, with occupation functions interpreted in the appropriate operator order,

$$\mathcal{L} = \sum_{i=1}^{d} \left[(a_i^\dagger - I)b_i(\widehat{\boldsymbol{N}}) + \mu_i(I - a_i^\dagger)a_i\right], \qquad \frac{d}{dt}|P\rangle = \mathcal{L}|P\rangle. \tag{7}$$

This is not a unitary quantum Hamiltonian: $|P\rangle$ contains classical probabilities. Conservation of probability mass is expressed by the summation functional $\langle 1|\mathcal{L} = 0$. This is standard Doi–Peliti theory [1,2].

3.2 Why modes interact

If a distribution is expanded in a modal basis, $|P\rangle = \sum_\alpha c_\alpha|\phi_\alpha\rangle$, a formal CME projection yields

$$\dot{c}_\alpha = \sum_\beta A_{\alpha\beta}c_\beta, \qquad A_{\alpha\beta} = \langle\phi_\alpha|\mathcal{L}|\phi_\beta\rangle, \tag{8}$$

with the appropriate projection metric. Hill functions and birth–death transitions couple different orders and species. In general, no exact closure exists in a small number of modes. Such couplings arise from the physical generator, not from a matrix fitted retrospectively to the solution.

3.3 Memory from discarded modes

For a linear modal equation, separate retained coefficients u from discarded coefficients v:

$$\dot{u} = A_{ll}u + A_{lh}v, \qquad \dot{v} = A_{hl}u + A_{hh}v. \tag{9}$$

Eliminating v exactly gives

$$\dot{u}(t) = A_{ll}u(t) + A_{lh}e^{A_{hh}t}v(0) + \int_0^t A_{lh}e^{A_{hh}(t-s)}A_{hl}u(s)\,ds. \tag{10}$$

This explains why higher-order modes exert memory effects on lower-order modes. The identity is a linear instance of Mori–Zwanzig and is not itself new. An auxiliary memory embedding of this kind was included in the tests; it was neither the winning variant nor a solution to positivity.

4 Constructing probability correctly

4.1 Option I: linear projection (a potentially invalid control)

For an orthonormal mode matrix B, a linear approximation $\tilde{p} = Bc$ gives $\dot{c} = B^\top QBc$. It is inexpensive and transparent, but can yield $\tilde{p}_n < 0$. Clipping negative entries and renormalizing only for evaluation does not turn the underlying evolution into a valid probabilistic CME. This is explicitly labeled a non-positive method in the results.

4.2 Option II: positive squared amplitude

The principal proposal tested is

$$\Psi_n(t) = \sum_{\alpha=1}^{K} B_{n\alpha}c_\alpha(t), \qquad P_n(t) = \frac{\Psi_n(t)^2}{\sum_m \Psi_m(t)^2}. \tag{11}$$

If the columns of B are orthonormal and $\|c\|_2 = 1$, the denominator equals one. The probability density is nonnegative and normalized, regardless of the signs of Ψ. This statistical amplitude should not be confused with the Doi–Peliti probability vector: the representations are related but distinct. Wherever $\Psi_n \neq 0$, exact CME dynamics imply

$$\dot{\Psi}_n = \frac{(Q\Psi^2)_n}{2\Psi_n}, \qquad \dot{c}_\alpha = \frac{1}{2}\sum_n B_{n\alpha}\frac{(Q\Psi^2)_n}{\Psi_n}. \tag{12}$$

The implementation approximates $1/\Psi \approx \Psi/(\Psi^2+\epsilon)$ and projects the velocity onto the unit sphere $\|c\| = 1$. This is an explicit numerical regularization: the resulting reduced evolution is not exactly the CME. Heun integration is used, varying both ϵ and the maximum time step.

4.3 Option III: exponential/entropy family

Another positive representation is

$$P_\theta(\boldsymbol{n}) = \frac{\rho(\boldsymbol{n})\exp(\sum_j \theta_j g_j(\boldsymbol{n}))}{Z(\theta)}, \quad Z(\theta) = \sum_{\boldsymbol{n}} \rho(\boldsymbol{n}) e^{\sum_j \theta_j g_j(\boldsymbol{n})}. \tag{13}$$

Enforcing the CME evolution of observable moments, $d\mathbb{E}_\theta[g]/dt = \mathbb{E}_\theta[Q^\top g]$, gives

$$\mathrm{Cov}_\theta(g,g)\,\dot{\theta} = \mathbb{E}_\theta[Q^\top g]. \tag{14}$$

This is an information-projection closure with Fisher geometry. It guarantees positivity and conservation of mass but not that the chosen family contains the important multimodal states.

4.4 Option IV: positive mixture of coherent reference states

Consider a dictionary $\rho_j(\boldsymbol{n}) \geq 0$ with $\sum_{\boldsymbol{n}} \rho_j(\boldsymbol{n}) = 1$ and $P \approx \sum_j w_j \rho_j$, where $w_j \geq 0$, $\sum_j w_j = 1$. A reduced generator K is fitted by least squares to $Q\rho_j$, subject to

$$K_{ij} \geq 0 \quad (i \neq j), \qquad \sum_i K_{ij} = 0, \qquad \dot{w} = Kw. \tag{15}$$

Once K has been constructed, this Markovian mixture can evolve without rebuilding the full microstate distribution. Its accuracy in our networks was poorer than the multi-reference amplitude, exposing a genuine tension between strictly reduced dynamics and accuracy.

5 The decisive element: multi-reference Fock bases and coherent shifts

5.1 Why a single center is insufficient

A bistable system can simultaneously assign high probability to states near $\boldsymbol{n} \approx 0$ and to high-occupancy states. A polynomial basis orthogonal about only one reference measure requires many degrees to capture both regions, wasting modal capacity on the low-probability gap between peaks.

Introduce several reference centers ρ_s. For the bounded occupation numbers considered, use a product of binomial distributions:

$$\rho_s(\boldsymbol{n}) = \prod_{i=1}^{d} \binom{M}{n_i} \theta_{s,i}^{n_i}(1-\theta_{s,i})^{M-n_i}. \tag{16}$$

These are natural product measures in a finite occupation space, not literally the Poissonian bosonic coherent states of infinite-dimensional Doi–Peliti theory. This distinction is

intentional. For each center we construct discrete orthonormal polynomials $q_k^{(s,i)}(n_i)$ with binomial weight, and tensor-product modes

$$\phi_{s,\alpha}(\boldsymbol{n}) = \prod_{i=1}^{d} \sqrt{\rho_{s,i}(n_i)}\, q_{\alpha_i}^{(s,i)}(n_i), \qquad |\alpha| = \sum_i \alpha_i \le K. \tag{17}$$

The combined dictionary is orthogonalized through the spectrum of its Gram matrix. Numerically dependent columns are removed according to a code-specified threshold. This produces the multi-reference basis B used in the squared-amplitude model (11).

5.2 Selecting centers without peeking into the future

For each component of the initial distribution, approximate centers $\theta_{s,i}(t)$ are propagated using mean-field equations derived from the same chemical rates:

$$M\dot{\theta}_{s,i} = \mathbb{E}_{\rho_s}[b_i(\boldsymbol{N}) - \mu_i N_i]. \tag{18}$$

Centers are taken at $t = 0, T/2, T$ using only the CME and the initial state, without exact reference distributions at later times. Amplitudes are then integrated according to (12). This is the best-performing construction in the study. Higher modes contain cross-species products (correlations) as well as deformations of individual marginals. Representation capacity generally increases with total degree, though a maximum-column restriction may destroy monotonicity.

5.3 Structural scalability limitation

The explicit basis contains $L^d \times K$ entries; moreover, evaluating $Q(\Psi^2)$ traverses the full probability vector. Even if a state is described by K coefficients, the implementation does not have complexity polynomial only in d. A genuinely scalable method would need factorizations of the generator and modal contractions, or an effective approximation to $(Q\Psi^2)/\Psi$ that preserves positivity.

6 Experimental design and competing approaches

6.1 Five scenarios with increasing complexity

Table 1: Test reaction networks.

Scenario	Species	Microstates	Main challenge
Independent	4	2,401	Negative control: mean field should be exact
Weak coupling	4	2,401	Small correlations and modal convergence
Bistability	5	7,776	Initial mixture of distant states
Regulatory cascade	6	46,656	Activation and inhibition across species
Strong coupling	7	279,936	Greater dimension, stiffness, alternating dependencies

Each reference distribution is computed as $e^{Qt}P(0)$ with SciPy's `expm_multiply`, evaluating $t = 0, T/4, T/2, 3T/4, T$. Reported mean subsequent errors exclude the initial instant; initial errors are also given to identify representation bias.

6.2 Relevant competitors

1. **Mean field:** integration of one-dimensional marginals; in mixtures each component follows its own trajectory.
2. **Matrix DLR with a species partition:** represent a rank-r matrix as $P \approx USV^\top$ and integrate its tangent projector via a K–S–L splitting. Apply the generator as $\sum_k A_k P B_k^\top$, without storing the full vector during evolution. This is legitimate predictive reduction, though not a state-of-the-art hierarchical tensor train.

3. **Elementary adaptive finite-state projection (FSP):** retain states deemed most probable by the mean-field approximation rather than the exact solution. Mass leakage is reported; large leakage means the conditioned approximation is not a certified FSP solution.
4. **Full CME with Krylov:** an error reference and timing baseline. Dimensional compression must not be confused with actual speedup.

The literature already includes more sophisticated algorithms, including certified FSP and hierarchical tensor reductions for 20–41 species [3–6]. This study does not reproduce their full implementations. It makes a reproducible comparison against strong matrix DLR and less sophisticated controls.

6.3 Evaluation metrics

$$E_{L^1}(t) = \sum_{\boldsymbol{n}} |P_{\boldsymbol{n}}(t) - \widehat{P}_{\boldsymbol{n}}(t)|, \qquad \mathrm{TV}(P, \widehat{P}) = E_{L^1}/2. \tag{19}$$

$$H(P, \widehat{P}) = \left[\frac{1}{2} \sum_{\boldsymbol{n}} \left(\sqrt{P_{\boldsymbol{n}}} - \sqrt{\widehat{P}_{\boldsymbol{n}}} \right)^2 \right]^{1/2}. \tag{20}$$

$$E_{\mathrm{event}}(t) = \Pr(N_1 \geq \lceil M/2 \rceil, N_2 \geq \lceil M/2 \rceil) - \widehat{\Pr}(\cdots). \tag{21}$$

We also measure error in $\mathbb{E}[N_1 N_2]/M^2$, negative mass before any correction, parameter count, and run time. For two valid distributions, $E_{L^1} \in [0, 2]$.

7 Independent verification of the generator

Two numerically independent representations were implemented: a sparse Q assembled from explicit transitions, and a sum of Kronecker products of local operators $\sum_k A_k \otimes B_k$. The maximum disagreement on a test vector was on the order of 10^{-18}; column-sum and normalization errors were at machine precision. The code includes a repeatable audit procedure.

An implementation error affecting degradation transitions was found in an earlier version. All reported figures are from the corrected implementation exclusively. This verification is crucial, because a wrong generator may make solvers appear superior when they actually solve different equations.

8 Main results: five networks

Table 2: Mean L^1 error at four times after the initial instant. Fock is the best tested multi-reference configuration for each network; DLR uses no future solution information.

Scenario	States	Mean field	Fock modes	Fock L^1	DLR $r = 2$	DLR $r = 4$
Independent, 4	2,401	0.00000	14	0.00003	0.00004	0.00004
Weak, 4	2,401	0.0215	86	0.00196	0.0103	0.00019
Bistable, 5	7,776	0.1221	400	0.0326	0.0652	0.0162
Cascade, 6	46,656	0.3333	360	0.0394	0.0681	0.00955
Coupled, 7	279,936	0.1438	215	0.0807	0.0731	0.0200

In bistability and the cascade, Fock achieves less error with a smaller representation dimension than rank-2 DLR. Rank-4 DLR beats Fock in every nontrivial case, at the cost of additional parameters. In the seven-species system, Fock (215 modes) approaches the error of rank-2 DLR (3,020 parameters), but does not beat it.

9 More modes: when do they help and when do they stop helping?

With five species, degree 4 and 400 modes yields 0.0326. When degree 5 was attempted but only 450 columns were retained out of 1,512 candidates, error increased to 0.1161 and the initial representation worsened sharply. This does not show that degree 5 is intrinsically worse: spectral truncation of the larger Gram matrix does not preserve the earlier modes as a nested subspace. Modal selection matters as much as modal count.

For seven species, constructing a still larger basis (degree 3, 720 candidates) exhausted available memory before completion. No valid numerical result exists for that attempt, and no trend is extrapolated from it.

10 What each probability construction contributes

The exponential family guarantees positivity, but the selected sufficient statistics do not adequately represent mixtures of regimes. The linear ansatz is often less accurate and can produce negative probability mass. The Markovian dictionary mixture preserves probabilistic constraints, yet its least-squares reduced generator loses some of the distribution's evolution.

Table 4: Selected formulations evaluated with identical metrics. Complete configurations are stored in `master_results.csv`.

System	Method	Coefficients	L^1
Weak, 4	Mean field	28	0.0215
Weak, 4	Linear Fock, degree 3	35	0.4169
Weak, 4	Single-center root, degree 3	35	0.1629
Weak, 4	Exponential, degree 2	14	0.00472
Weak, 4	Valid Markov mixture	15	0.0738
Weak, 4	Rank-2 DLR	192	0.0103
Bistable, 5	Mean field	61	0.1221
Bistable, 5	Linear Fock, degree 3	56	1.1131
Bistable, 5	Single-center root, degree 3	56	1.0696
Bistable, 5	Exponential, degree 2	20	0.6286
Bistable, 5	Multi-reference root, degree 4	400	0.0326
Bistable, 5	Valid Markov mixture	36	0.4269
Bistable, 5	Rank-2 DLR	500	0.0652
Cascade, 6	Mean field	36	0.3333
Cascade, 6	Linear Fock, degree 3	84	0.5807
Cascade, 6	Single-center root, degree 3	84	0.9040
Cascade, 6	Multi-reference root, degree 3	245	0.0476
Cascade, 6	Valid Markov mixture	21	0.5512
Cascade, 6	Rank-2 DLR	860	0.0681
Coupled, 7	Mean field	85	0.1438
Coupled, 7	Linear Fock, degree 3	120	1.0223
Coupled, 7	Single-center root, degree 3	120	0.9709
Coupled, 7	Multi-reference root, degree 2	215	0.0807
Coupled, 7	Valid Markov mixture	48	0.4226
Coupled, 7	Rank-2 DLR	3,020	0.0731

11 Matching moments is not enough: correlations and events

The most convincing advantage of the multi-reference representation is not fitting one mean or variance: it captures non-Gaussian joint distributions with relatively few coefficients. Nevertheless, the full-distribution E_{L^1} is the principal benchmark, preventing method selection based only on an observable chosen after seeing the results.

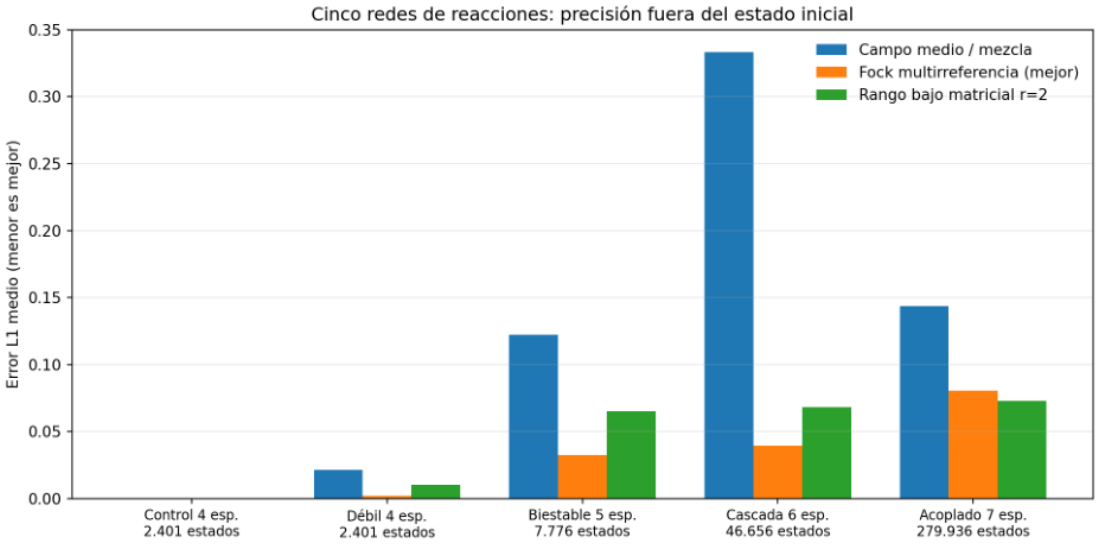


Figure 1: Comparison against two competitors on identical networks and time points. The independent control should be solved almost exactly by mean-field closure. Original plot labels are retained from the source figure.

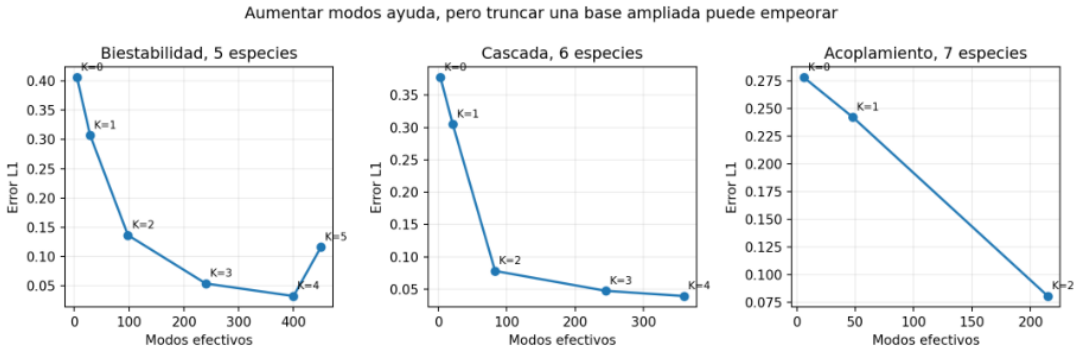


Figure 2: Error versus effective mode count. Polynomial degree is the maximum total order over several centers; high-degree cases have a column cap and the bases are not perfectly nested. Original plot labels are retained.

Table 3: Multi-reference expansions: initial and subsequent errors. Effective mode count follows the removal of Gram redundancies; expanded experiments impose a maximum number of coefficients.

System	Degree	Modes	$L^1(t=0)$	Later L^1	Event error
Bistable, 5	0	5	0.00123	0.4056	0.0214
Bistable, 5	1	29	0.00083	0.3071	0.0214
Bistable, 5	2	98	0.00050	0.1365	0.00121
Bistable, 5	3	240	0.00052	0.0540	0.00029
Bistable, 5	4	400	0.00055	0.0326	0.00050
Bistable, 5	5	450	0.0313	0.1161	0.00384
Cascade, 6	0	3	0.00000	0.3771	0.0288
Cascade, 6	1	21	0.00000	0.3052	0.00641
Cascade, 6	2	83	0.00000	0.0783	0.00303
Cascade, 6	3	245	0.00000	0.0476	0.00045
Cascade, 6	4	360	0.0295	0.0394	0.00259
Coupled, 7	0	6	0.00003	0.2779	0.00949
Coupled, 7	1	48	0.00003	0.2420	0.0102
Coupled, 7	2	215	0.00002	0.0807	0.00212

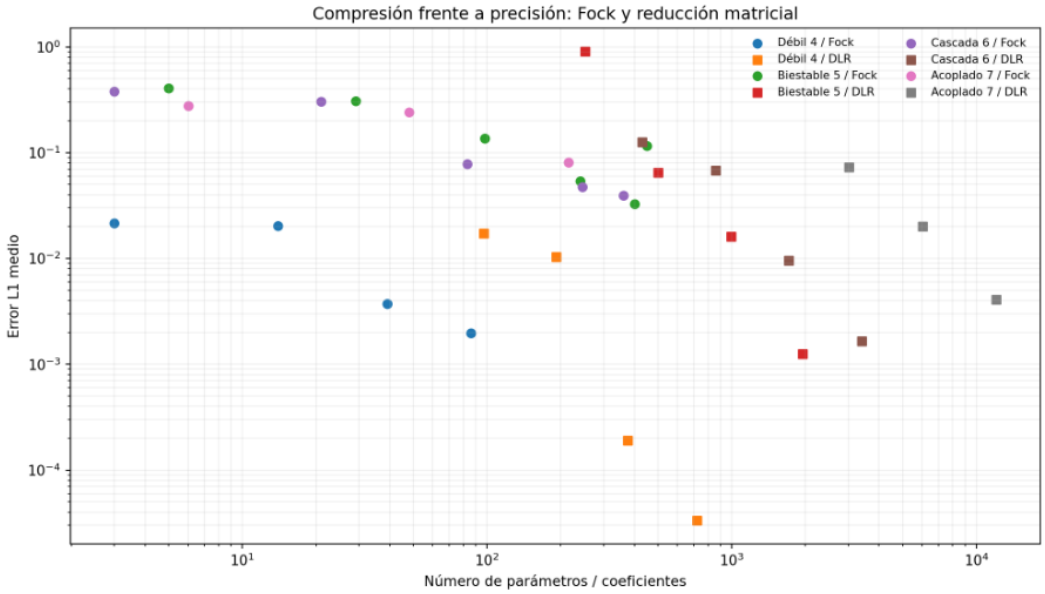


Figure 3: Empirical complexity frontier: parameter count versus distribution error. Circles mark multi-reference Fock results and squares mark matrix DLR. A unique winner cannot be declared without specifying memory and accuracy budgets. Original plot labels are retained.

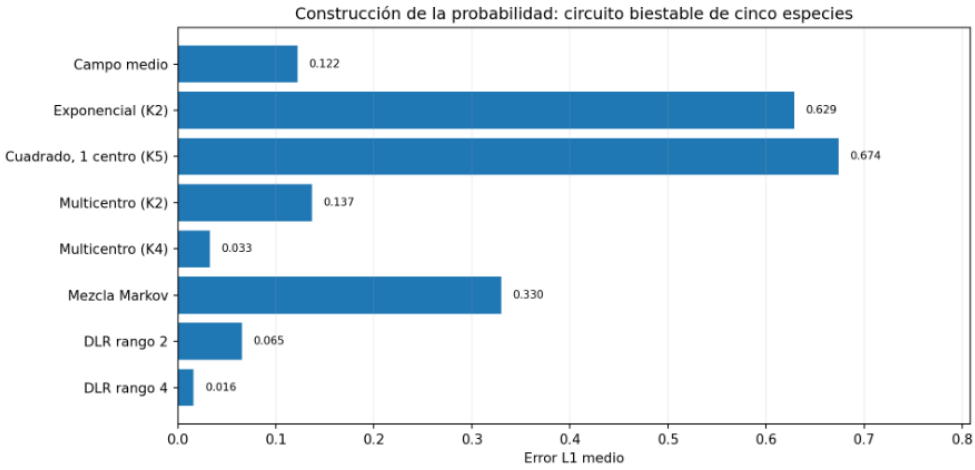


Figure 4: Five-species bistable circuit. Increasing the number of reference centers is more valuable than raising polynomial degree around one center. Positive Markov closure is stable but too restrictive in this configuration. Original plot labels are retained.

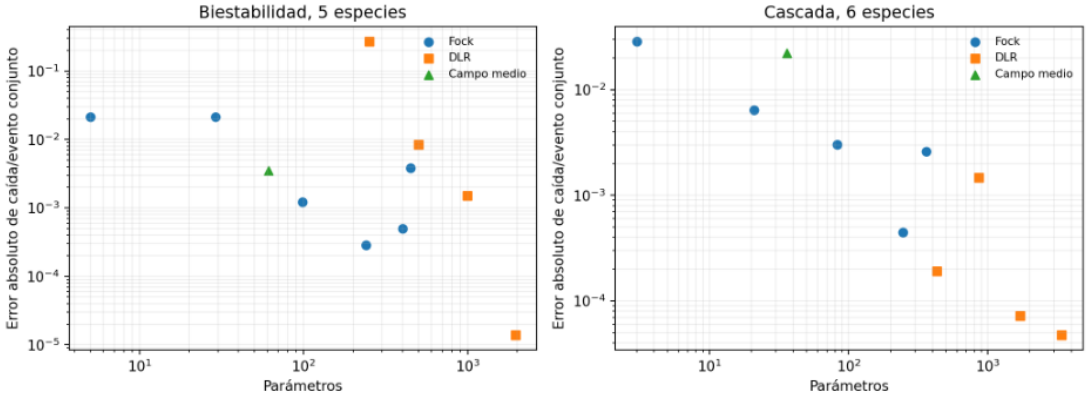


Figure 5: Absolute error in the probability that the first two species simultaneously have high occupation. A correct distribution yields correct observables, but the converse does not hold. Original plot labels are retained.

12 Numerical sensitivity and robustness checks

The dimensionless regularization strength of the amplitude division and the maximum integration step were varied as follows:

$$\epsilon/(1/N) \in \{10^{-3}, 10^{-5}, 10^{-7}\}, \qquad \Delta t \in \{0.0125, 0.025, 0.05\}.$$

Two representative configurations were completed across a 3×3 grid: An important caveat:

Table 5: Range of errors under changes in regularization and integration step.

Case and modes	Tests	Min.	Median	Max.
Bistable, 400	9	0.0325	0.0326	0.0327
Cascade, 245	9	0.0474	0.0475	0.0477

these are synthetic systems with known mechanisms and fixed parameters. Robustness to systematic kinetic-law misspecification, changes in the support size, and error in estimating parameters from experiments requires separate study.

Table 6: Measured single-thread execution time (seconds). For the expanded Fock variant, basis construction and evolution are included; for the reference, propagation is timed without initial assembly of Q.

System	Method	L^1	Time (s)
Bistable, 5	Fock (400)	0.0326	0.490
	DLR rank 2	0.0652	0.073
	Full CME Krylov	0	0.009
Cascade, 6	Fock (245)	0.0476	1.570
	DLR rank 2	0.0681	0.121
	Full CME Krylov	0	0.060
Coupled, 7	Fock (215)	0.0807	4.620
	DLR rank 2	0.0731	0.135
	Full CME Krylov	0	0.465

13 Actual cost: compression versus a practical algorithm

It must not be overlooked that full CME propagation via Krylov is faster at the sizes examined. DLR is also genuinely reduced *during* evolution: it uses separable products of chemical operators without reconstructing the full matrix. The current Fock method computes Qp in dimension L^d. Consequently:

- **Observed advantage:** very few coefficients achieve moderate error on correlated systems.
- **Advantage not observed:** lower run time or working memory for solving the CME.
- **Outstanding objective:** evaluate modal couplings and the nonlinear amplitude division using tensor contractions that avoid building a vector with L^d entries.

The elementary FSP based on mean-field-selected states suffered substantial mass leakage in highly coupled cases. It is therefore neither a certified FSP baseline nor directly competitive with modern adaptive methods [3]. Detailed leakage figures are included in a separate CSV.

14 What is original, what is not, and what warrants further work

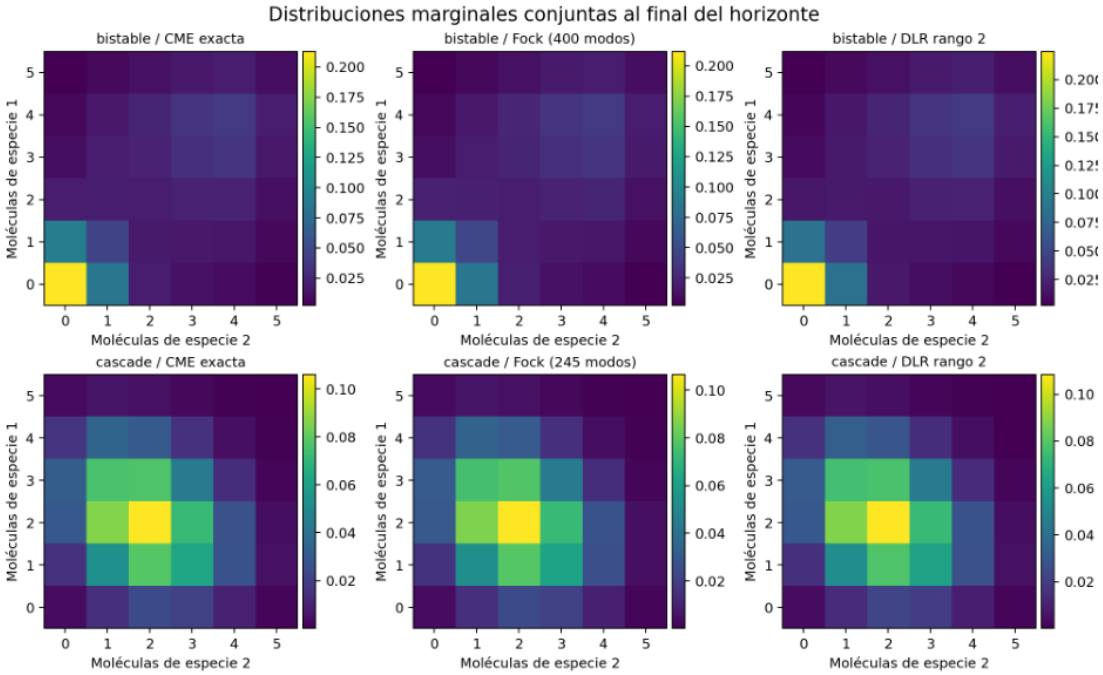


Figure 6: Final-time joint marginals of (N_1, N_2) from the full solver, multi-reference Fock, and rank-2 DLR. These are predictions, not retrospective fits to the reference. Original plot labels are retained.

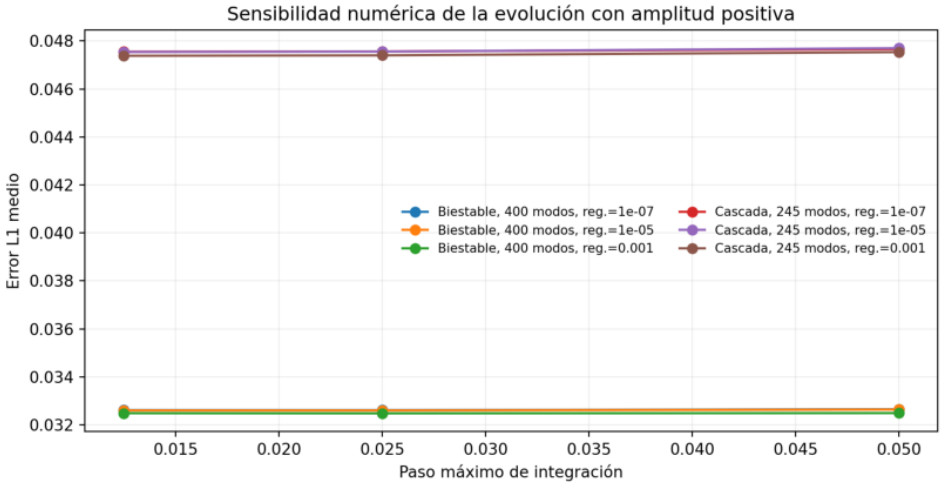


Figure 7: Integrator variations for both completed sensitivity tests are much smaller than differences between model families. The full sensitivity test for seven species was not completed and cannot be counted as universal validation. Original plot labels are retained.

14.1 The most promising observation

The form $\Psi = \sum_{s,\alpha} c_{s,\alpha}\phi_{s,\alpha}$, with centers shifted using an inexpensive physically motivated prediction and $P \propto \Psi^2$, is a defensible working hypothesis for compact representation of multimodal stochastic distributions. The improvement over a single-center expansion is substantial in correlated regulatory circuits. Positivity does not require ad hoc post-hoc clipping.

14.2 What has not been discovered

Fock occupation representations, coherent states, Galerkin projection, the Mori–Zwanzig equation, information closures, and tensor networks are established results. The tested multi-reference combination is an exploratory numerical implementation. No theorem establishing a convergence rate has been proved, and novelty over the entire literature has not been demonstrated.

A 2025 study applies DMRG to a seven-species genetic switch [7]. Modern DLR approaches have tackled 20-dimensional systems and 41-species circuits [4,6]. A paper claiming superiority over them would need direct comparisons with their implementations and a demonstrated improvement on at least one relevant metric.

14.3 Conditions for a reasonable continuation

We propose just one additional stage, explicitly limited and with a stopping criterion:

1. Derive local or tensorial contraction rules for $B^\top[Q(\Psi^2)/\Psi]$ that avoid the full vector, and quantify any new error in positivity and mass conservation.
2. Make mode selection nested and adaptive to preserve the initial state when increasing polynomial degree.
3. Test a network with at least 10–20 species and published biological parameters, comparing against a modern tensor DLR and/or tensor-train implementation at fixed accuracy tolerance.
4. Measure total time including construction of modes and generators, and check rare-event transition probabilities and tail errors.

If no advantage emerges in cost, stability, or error control over those methods, continuing this line as a new algorithm would not be justified.

15 Final assessment

Does it work as an approximation? Yes. A multi-center Fock construction with squared probability amplitudes captures joint distributions of moderately sized nonlinear biochemical networks using few coefficients.

Is it globally better than current methods? No. Low-rank DLR provides better maximum accuracy, much faster reduced evolution, and superior scalability in the evaluated scenarios.

Is further work worthwhile? Only conditionally. The positive results justify documenting the project as a mathematical exploration and, optionally, attempting one improvement in modal contraction and adaptive closure. Claims of a superior theory or applied biological utility would be premature.

16 Complete reproduction procedure

The accompanying ZIP, if supplied separately, contains original experiment scripts and outputs rather than manually reconstructed plots:

```
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python experiment.py --case 4
OPENBLAS_NUM_THREADS=1 python experiment.py --case 5
OPENBLAS_NUM_THREADS=1 python experiment.py --case 6
OPENBLAS_NUM_THREADS=1 python experiment.py --case 7
OPENBLAS_NUM_THREADS=1 python extended.py --case 5
OPENBLAS_NUM_THREADS=1 python extended.py --case 6
OPENBLAS_NUM_THREADS=1 python extended.py --case 7
OPENBLAS_NUM_THREADS=1 python fsp_control.py
OPENBLAS_NUM_THREADS=1 python aggregate_figures.py
```

For high-degree variants in the seven-species network, more memory is recommended than was available in the test environment; the degree-3 attempt was not completed and is omitted from the numerical results. Sensitivity scripts allow each scenario to be run independently and lengthy experiments to be interrupted without losing earlier data. CSV fields include `L1_t0`, `raw_negative`, `parameters`, `seconds`, and event errors.

Hardware and reproducibility: timings come from individual runs in a Linux container with single-threaded BLAS. They are indicative, not a formal performance study across hardware architectures. A random-number generator is used only in parity tests, not when propagating the five scenarios.

17 Background bibliography

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[4] Prugger, M., Einkemmer, L. and Lopez, C. F. (2023). *A dynamical low-rank approach to solve the chemical master equation for biological reaction networks*. J. Comput. Phys. 489, 112250. DOI: 10.1016/j.jcp.2023.112250.

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[6] Einkemmer, L., Mangott, J. and Prugger, M. (2026). *A hierarchical dynamical low-rank algorithm for the stochastic description of large reaction networks*. SIAM J. Sci. Comput. DOI: 10.1137/24M1685717.

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A Methodological audit notes

1. **Finite systems.** Molecular occupation is bounded by design; capacity factors belong to the network definition, rather than being a later approximation to an unbounded process.
2. **Shifted bases.** Centers come from initial conditions and mean-field closure, not future exact distributions. Construction of the Gram matrix still traverses all states, so the approach is not yet scalable.
3. **Fair comparison.** DLR uses the same factored chemical generator and is not fitted to the reference. Its parameter count is the dimension of a rank-r matrix manifold, approximately $r(m + n - r)$.

4. **DLR positivity.** Low-rank matrix projection may produce negative entries. The CSV stores negative mass before output clipping. Reconstructed squared-amplitude Fock distributions always have zero negative mass.
5. **Non-nestedness.** Adding a polynomial degree and then capping columns via the Gram spectrum may discard previously included directions. Oscillating errors should not be interpreted as an asymptotic convergence law for the theory.
6. **Numerical reference.** Full CME evolution uses Krylov without Monte Carlo trajectories or sampling noise. This reference solution is not empirical observation.
7. **Scope of "exhaustive".** The listed variants are audited over five representative reaction networks; this does not exhaust all possible operators or biological networks, and no universal superiority result is claimed.